
E-Commerce Customer Segmentation Analysis

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Code & Documentation : [GitHub Repository](#)

Executive Summary

A multi-category E-Commerce business faced a challenge with customer retention—specifically, a "Leaky Bucket" scenario where high-value customers were becoming inactive. The objective of this analysis was to segment the customer base to identify these at-risk groups and develop targeted retention strategies.

Using **RFM Analysis (Recency, Frequency, Monetary)** and **K-Means Clustering** on 2.24 million transaction records, the customer base was divided into four distinct strategic segments. The analysis successfully identified the top tier "Champions" (High LTV) and a critical "At-Risk" segment, creating a roadmap to maximize revenue through personalized marketing.

Methodology

The analysis followed a rigorous data science pipeline to ensure statistical validity and business relevance:

1. **Data Pre-processing:** From an initial dataset of 2.24 million records, the data was filtered to isolate valid "Purchase" transactions, resulting in a clean dataset of over 100,000 unique customer journeys.
2. **RFM Calculation:** Customer behavior was quantified using Recency (days since last purchase), Frequency (total transactions), and Monetary value (total spend).
3. **Statistical Transformation:** To address the heavy skew in spending data and ensure algorithmic stability, **Log Transformation** and **StandardScaler** normalization were applied.
4. **Model Selection:** The **K-Means Clustering** algorithm was selected. While the Silhouette Score suggested a mathematical preference for 2 clusters, the **Elbow Method** and business domain logic confirmed that **K=4** provided the most actionable distinction between new, loyal, and churning customers.

Data Source:

The analysis utilized the 'eCommerce Events History in Cosmetics Shop' dataset, provided by Michael Kechinov from Kaggle.

Link: [Kaggle - eCommerce Events History](#)

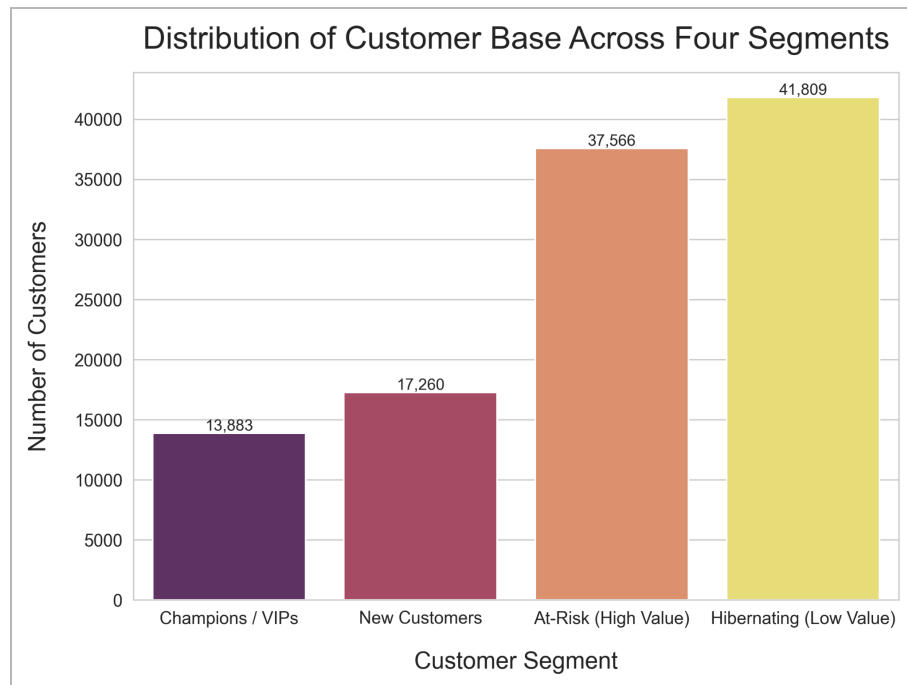
NOTE: Despite the dataset title, exploratory data analysis revealed a multi-category product mix including electronics and appliances, suggesting a general department store environment rather than a niche cosmetics shop.

Strategic Segmentation & Recommendations

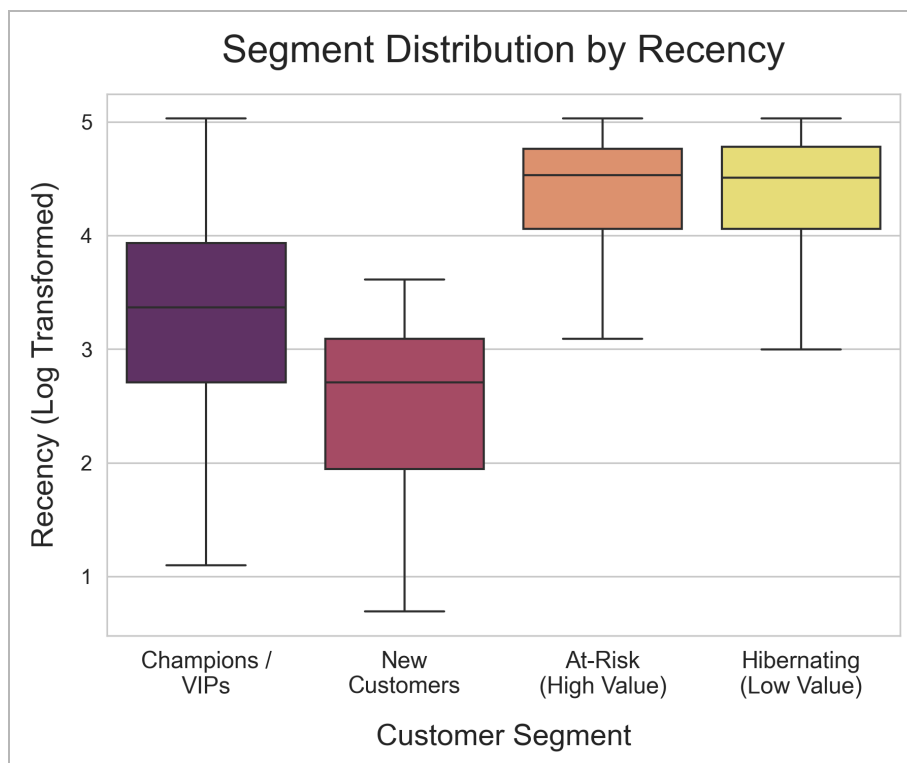
The analysis identified four customer segments. The table below outlines the recommended strategy to maximize the Lifetime Value (LTV) of each group:

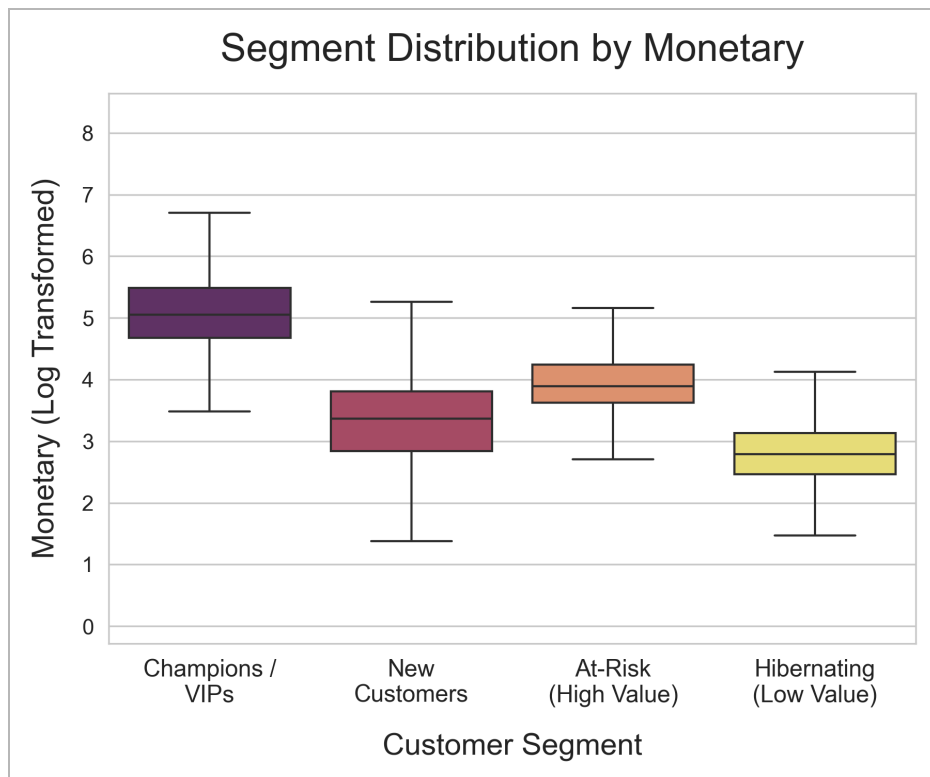
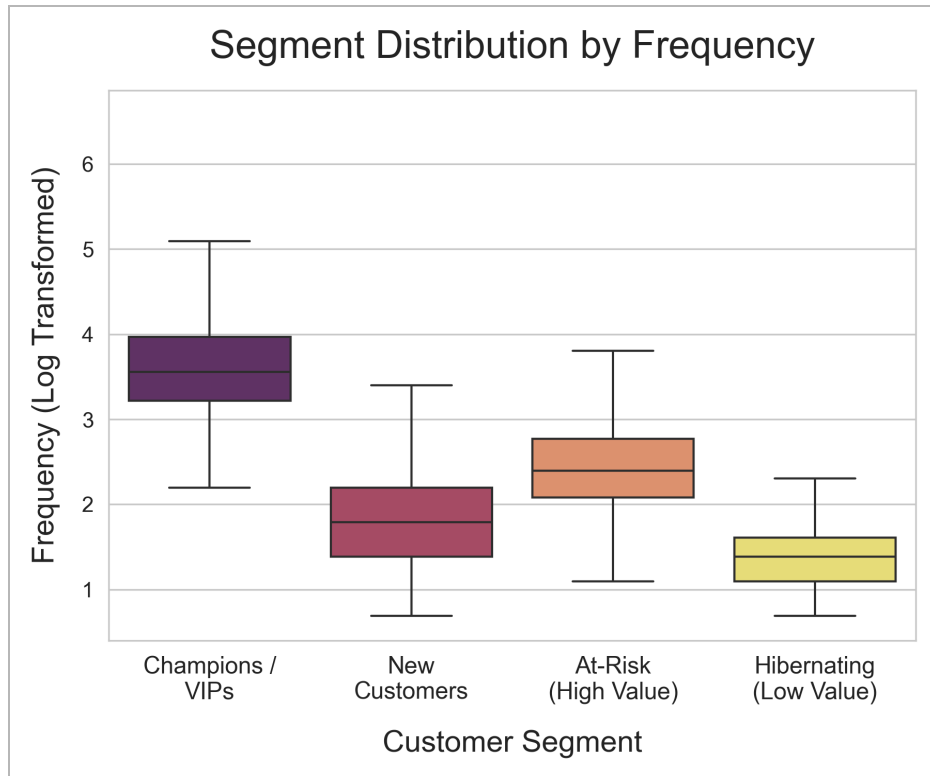
Segment	Profile	Strategy	Actionable Tactic
Champions / VIPs	High Spend, Recent, Frequent	Retention & Loyalty	Launch a "Surprise & Delight" program with exclusive access and non-monetary rewards to lock in loyalty.
New Customers	Recent, Low Frequency	Onboarding & Growth	Trigger a "Welcome Series" with a time-limited incentive (e.g., 10% off) to drive the critical second purchase.
At-Risk (High Value)	High Past Spend, Inactive	Aggressive Win-Back	Deploy high-value offers (20-30% off) targeted specifically at their preferred product categories to regain attention.
Hibernating	Low Spend, Long Inactive	Cost Avoidance	Test a final, low-cost automated email. If no response is received within 90 days, remove from ad targeting to save budget.

The following visualizations confirm the distinct behavioral profiles of the four identified segments across the Log-transformed RFM metrics, validating the cluster selection.



Distribution of the customer base. Note the large volume of Hibernating customers vs. the high-value Champion segment.





Behavioral profiles of each segment. Champions (Purple) show distinctively higher Monetary value and Frequency.

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